FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 15-12-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. Amplitude of two waves at point 'P' is $\frac{C}{r_1}$ & $\frac{C}{r_2}$ respectively, where 'C' is a constant

Let, $y_1 = \frac{C}{r_1}\sin(kr_1 - \omega t)$

$y_2 = \frac{C}{r_2}\sin(kr_2 - \omega t)$

$y = y_1 + y_2$

$= \frac{C}{r}\left[\sin(kr_1 - \omega t) + \sin(kr_2 - \omega t)\right]$

$= \frac{2C}{r}\cos\frac{k}{2}(r_1 - r_2).\sin(kr - \omega t)$

2. C

Sol. $\theta_1 = \theta_0 \sin(\omega_1 t + \pi/2)$

$\theta_2 = \theta_0 \sin(\omega_2 t + 3\pi/2)$

For $\theta_1 = \theta_2$

$\cos\omega_1 t = -\cos\omega_2 t$

$\cos(\omega_1 t) = \cos(\pi - \omega_2 t)$

$(\omega_1 + \omega_2)t = \pi$

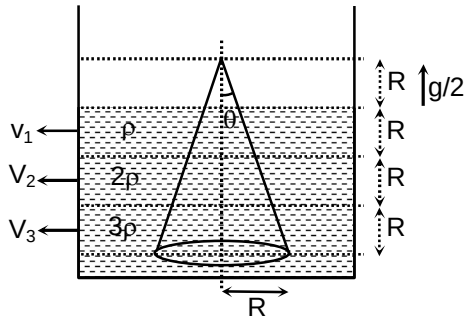
$2\left(\frac{1}{T_1} + \frac{1}{T_2}\right)t = 1$

Where $T_1 = 2\pi\sqrt{\frac{L_0}{g}}$, $T_2 = 2\pi\sqrt{\frac{25L_0}{g}}$

$t = \frac{5\pi}{6}\sqrt{\frac{L_0}{g}}$

3. C

Sol. $\text{Tan}\,\theta = \frac{1}{4}$



$v_1 = \frac{\pi}{3}\left[x_2^2.2R - x_1^2.R\right] = \frac{7\pi}{48}R^3$

$v_2 = \frac{\pi}{3}\left[x_3^2.3R - x_2^2.2R\right] = \frac{19\pi}{48}R^3$

$v_3 = \frac{\pi}{3}\left[R^2.4R - x_3^2.3R\right] = \frac{37\pi}{48}R^3$

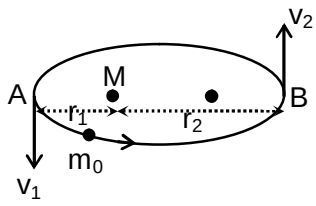
So, $m = v_1\rho + v_2(2\rho) + v_3(3\rho)$

$\rho_0\frac{\pi R^2}{3}\times 4R\left(\frac{3g}{2}\right) = \rho\frac{\pi R^3}{48}(156)\left(\frac{3g}{2}\right)$

Density of cone, $\rho_0 = \frac{39}{16}\rho$

4. D

Sol. $v_1r_1 = v_2r_2$ …(1)



$\frac{GMm_0}{r_1^2} = \frac{m_0v_1^2}{R_1}$

$\frac{GMm_0}{r_2^2} = \frac{m_0v_2^2}{R_2}$

So, $\frac{R_1}{R_2} = \frac{v_1^2\ r_1^2}{v_2^2\ r_2^2} = 1$

5. B, D

Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{dv}{dt} = -\frac{v^2}{u^2}\frac{du}{dt} = -8$ m/s

$m_T = -\frac{v}{u} = \frac{f}{f-u}$

$h_i = \frac{f}{f-u}h_0$

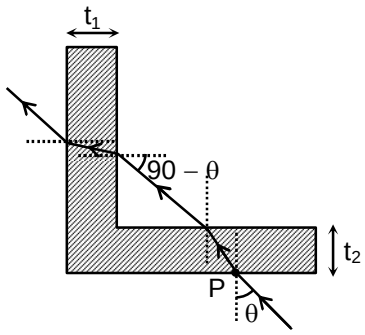
$\frac{dh_i}{dt} = \frac{fh_0}{(f-u)^2}\frac{du}{dt} = \frac{-20\times 0.2\times 2}{100} = \frac{-2}{25}$ cm/s

6. B, D

Sol. Use the concepts of refraction and TIR.

7. C, D

Sol.



8. A, D

Sol. $a \propto \frac{z^3}{n^4}$

$T \propto \frac{n^3}{z^2}$

9. A, D

Sol. $\Delta E = \frac{hc}{\lambda}$

$\Delta E = \frac{hc}{\lambda(\text{in nm})}(\text{in eV})$

10. B, C

Sol. $I = \frac{ma^2}{6} + m\left(\frac{a}{\sqrt{2}}\right)^2 + m(\sqrt{2}a)^2 = \frac{8}{3}ma^2$

$d = \frac{a}{\sqrt{2}} + \frac{a}{2\sqrt{2}} = \frac{3a}{2\sqrt{2}}$

$T = 2\pi\sqrt{\frac{I}{2mgd}} = 2\pi\sqrt{\frac{8\sqrt{2}}{9}\frac{a}{g}}$

11. A, B, C, D

Sol. $Q = \Delta K$

and use basic concept

12. A, B, C

Sol. In equilibrium

$4 mg \sin\theta = kx_0$

$k = 120{,}000$ N/m

$f = \frac{1}{2\pi}\sqrt{\frac{k}{2m}} = \frac{\sqrt{60}}{2\pi}$ Hz

New amplitude = 20/2 = 10 cm

SECTION – C

13. 00016.00

Sol. For refraction only

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R} \qquad \ldots(i)$$

$v = -10$ cm

Differentiate equation (i) w.r.t. time

$$-\frac{\mu_2}{v^2}\frac{dv}{dt}+\frac{\mu_1}{u^2}\frac{du}{dt}=0$$

$$\frac{dv}{dt}=\frac{\mu_1}{\mu_2}\frac{v^2}{u^2}\frac{du}{dt}=\frac{4}{1\times 3}\times 1\times 6=8 \text{ m/s (towards left)}$$

The velocity of image formed after refraction is 8 m/s (towards left)

The velocity of image formed after reflection and then refraction = 8 m/s (towards right).

The relative velocity between two images formed = 16 m/s

14. 00001.62

Sol. Path difference at 'O' = $(\mu_1 - \mu_2)t = 4.5 \times 10^{-6}$ m

Hence phase difference $\phi = \frac{2\pi}{\lambda}\cdot\Delta r = 20\pi$

Now, $I_1 = \frac{I_0}{16}$ and $I_2 = \frac{I_0}{25}$

So, $I = \left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

$= 1.62$ W/m^2

15. 00001.88

Sol. Let 'x' is the displacement of block from its equilibrium position.

$$m\frac{d^2x}{dt^2}=-kx-\frac{kx}{21}$$

$$\frac{d^2x}{dt^2}=-\left(\frac{22k}{21m}\right)x$$

$$T=2\pi\sqrt{\frac{21m}{22k}}=1.88 \text{ sec}$$

16. 00008.00

Sol. $v_1=\sqrt{\frac{F}{\mu_1}}=25$ m/s

$v_2=\sqrt{\frac{F}{\mu_2}}=20$ m/s

$$A_r=\left(\frac{|v_2-v_1|}{v_2+v_1}\right)A_i=\frac{2}{9} \text{ mm}$$

$$A_t=\left(\frac{2v_2}{v_2+v_1}\right)A_i=\frac{16}{9} \text{ mm}$$

$$\therefore \frac{A_t}{A_r}=8$$

17. 00012.80

Sol. $T = 2\pi\sqrt{\frac{I}{mgd}}$

So $\frac{T_P}{T_q} = \sqrt{\frac{I_P}{d_P}.\frac{d_q}{I_q}}$

$\frac{6}{10} = \sqrt{\frac{16}{25}\cdot\frac{d_q}{d_p}}$

$\frac{d_q}{d_p} = \frac{9}{16}$ …(1)

$d_q + d_p = 20$ …(2)

$\Rightarrow d_p = \frac{64}{5} = 12.80$

18. 00002.50

Sol. $\eta A\frac{dv}{dy} = kx = mg$

$5\times 2\times 10^{-2}\times\frac{4\times 10^{-2}}{2\times 10^{-3}} = 80x$

$x = \frac{2}{80}m = 2.50\,cm$

Chemistry

PART – II

SECTION – A

19. A

Sol. Meq of I_2 = meq of $Na_2S_2O_3 = 60 \times 0.2 = 12$

Wt. of I_2 liberated $= \frac{12 \times E}{1000}$

Also, $W = \frac{EIt}{96500}$

$\therefore \frac{EIt}{96500} = \frac{12 \times E}{1000}$

$= \frac{I \times 193}{96500} = \frac{12}{1000}$

I = 6 amp.

20. B

Sol.

	$2N_2O_5$	$\longrightarrow$	$4NO_2$	+	O_2
$t = 0$	10		0		0
t	6		8		2

$[N_2O_5]$ left = 6 M.

Rate of reaction = $2 \times 10^{-5} \times 6 = 12 \times 10^{-5}$ M sec^{-1}

Rate of formation of $NO_2 = 4 \times 12 \times 10^{-5} = 4.8 \times 10^{-4}$ M sec^{-1}

21. A

Sol. $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$\frac{T_2}{300} = \left(\frac{40}{5}\right)^{\frac{5}{3}-1}$

$T_2 = 300 \times (8)^{2/3}$

$T_2 = 300 \times \left(2^3\right)^{2/3}$

T_2 = 1200 K.

22. B

Sol. $\frac{P^o - P^S}{P^o} = \frac{n_B}{n_B + n_A}$

$= \frac{100 - 80}{100} = \frac{n_B}{n_B + n_A}$

$= \frac{20}{100} = \frac{\frac{W_B}{60}}{\frac{W_B}{60} + \frac{90}{18}}$

$W_B = 75$ g

23. B, C

Sol. $[NiCl_4]^{2-}$ and $[MnCl_4]^{2-}$ are tetrahedral complexes.

24. A, C, D

Sol. N_2O is a neutral oxide.

25. B, C, D

Sol. ClO_2 does not dimerise.

$Cl_2O_6 + H_2O \longrightarrow HClO_3 + HClO_4$

$I_2O_5 + 5CO \longrightarrow 5CO_2 + I_2$

Liquid IF_5 self ionizes and therefore conducts electricity.

$2IF_5 \rightleftharpoons IF_4^+ + IF_6^-$

26. A, C, D

Sol. Hg_2^{2+} and Ag^+ give white ppt. with dil. HCl.

27. B, D

Sol. $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$

$2PH_3 + 3HgCl_2 \longrightarrow Hg_3P_2 + 6HCl$

28. A, B, C, D

Sol. All the reactions are correct. XeF_4 fluorinate Pt.

29. A, C, D

Sol. SnS is soluble in yellow ammonium sulphide.

30. A, C

Sol. In aluminothermic process Cr_2O_3 is reduced by Al.
In Van-Arkel process impure Ti is purified by reacting with I_2.

SECTION – C

31. 00032.84

Sol. $\pi V = \frac{W}{M}RT \qquad \pi = \frac{38}{76} = 0.5 \text{ atm}$

$M = \frac{W\ RT}{\pi\ V}$

$= \frac{0.2 \times 0.0821 \times 300 \times 1000}{300 \times 0.5}$

= 32.84

32. 00001.28

Sol. Initial meq. of $Cu^{2+} = 50 \times 1 \times 2 = 100$

Electricity passed = $0.965 \times 3600 = 3474$ C

No of meq of Cu^{2+} discharged $= \frac{3474 \times 1000}{96500} = 36$

$\therefore$ Meq of Cu^{2+} remained = 100 – 36 = 64

$\therefore N = \frac{64}{50} = 1.28$

33. 00083.75

Sol. $C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$

$\Delta H_r^o = \sum \Delta H_f^o(\text{Products}) - \sum \Delta H_f^o(\text{Reactants})$

$-1564.25 = \left[2 \times (-395) + 3 \times (-286)\right] - \left[\Delta H_f^o C_2H_6\right]$

$\Delta H_f^o C_2H_6 = -83.75 \text{ kJ mol}^{-1}$

34. 00046.06

Sol. $k = \frac{0.693}{13.86}$

$t = \frac{2.303}{k}\log\frac{10}{1}$

$= \frac{2.303 \times 13.86}{0.693}$

= 46.06

35. 00013.25

Sol. Meq of Na_2CO_3 in 25 ml = meq of $H_2SO_4 = 20 \times \frac{1}{10} = 2$

Meq of Na_2CO_3 in 250 ml = 20

$\therefore$ wt of $Na_2CO_3 = \frac{20 \times 53}{1000} = 1.06$ g

% of $Na_2CO_3 = \frac{1.06}{8} \times 100 = 13.25$

36. 00024.63

Sol. $Z = \frac{PV}{nRT}$

$\frac{V}{n} = \frac{ZRT}{P} = \frac{0.8 \times 0.0821 \times 300}{8}$

= 2.463 litre/mole

$\therefore$ Volume of 10 mol of the gas = 24.63 litre.

Mathematics

PART – III

SECTION – A

37. A

Sol. 1, r, r^2, ….. , r^{2m} are in G.P.

$1 + r^{2m} > 2.r^m$

$r + r^{2m-1} > 2.r^m$

$r^2 + r^{2m-2} > 2.r^m$

$r^3 + r^{2m-3} > 2.r^m$

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Hence $1 + r + r^2 + \ldots\ldots + r^{2m} > 2(r^m + r^m + \ldots. \text{ m times}) + r^m$

$\left(\frac{1-r^{2m+1}}{1-r}\right) > 2m.r^m + r^m = (2m + 1).r^m$

$(2m + 1)r^m.(1 - r) < 1 - r^{2m+1}$.

38. A

Sol. $\alpha + \beta = a$, $\alpha\beta = -(a + b) \Rightarrow (1 + \alpha)(1 + \beta) = (\alpha + \beta) + \alpha\beta + 1 = -(a + b) + a + 1 = (1 - b)$

$$\Rightarrow \frac{\alpha^2+2\alpha+1}{\alpha^2+2\alpha+b}+\frac{\beta^2+2\beta+1}{\beta^2+2\beta+b}=\frac{(\alpha+1)^2}{(\alpha+1)^2+(b-1)}+\frac{(\beta+1)^2}{(\beta+1)^2+(b-1)}$$

$$= \frac{(\alpha+1)^2}{(\alpha+1)(\alpha+1-1-\beta)}+\frac{(\beta+1)^2}{(\beta+1)(\beta+1-1-\alpha)} = \frac{\alpha+1}{\alpha-\beta}+\frac{\beta+1}{\beta-\alpha}=\frac{\alpha-\beta}{\alpha-\beta}=1$$

39. C

Sol. $$y=\sum_{r=1}^{n-1}\frac{(r+1)^2-r(n-r)}{(n-r)\,{}^nC_r} = \sum_{r=1}^{n-1}\left(\frac{(r+1)}{\frac{n-r}{r+1}\,{}^nC_r}-\frac{r}{{}^nC_r}\right) = \sum_{r=1}^{n-1}\left(\frac{r+1}{{}^nC_{r+1}}-\frac{r}{{}^nC_r}\right) = \frac{n}{{}^nC_n}-\frac{1}{{}^nC_1}=n-\frac{1}{n}$$

40. B

Sol. $\frac{4x^2}{1+4x^2} = y \Rightarrow \frac{y}{x}=\frac{4x}{1+4x^2}$ and $1 + 4x^2 \geq 4x \Rightarrow \frac{y}{x} \leq 1 \Rightarrow y \leq x$

Similarly, $x \leq z$ and $z \leq y$. Hence, $x = y = z$

41. A, C

Sol. $|a-b|^2-|1-\bar{a}b|^2=\left(|a|^2+|b|^2-2\text{Re}(\bar{a}b)\right)-\left(1+|a|^2|b|^2-2\text{Re}(\bar{a}\bar{b})\right) = -\left(1-|a|^2\right)\left(1-|b|^2\right)<0$

Hence, $|a - b| < |1-\bar{a}b| \Rightarrow \left|\frac{a-b}{1-\bar{a}b}\right|<1$

42. B, D

Sol. x, a_1, a_2, ….. , a_{2n}, y are in A.P.

$d = \left(\frac{y-x}{2n+1}\right)$

$(2n + 1) = \sum_{r=1}^{2n} a_r = \sum_{r=1}^{2n}(x + rd)$

$= 2nx + n(2n + 1)d = n(2x + y - x)$

$(2n + 1) = (x + y)n \Rightarrow n = \left(\frac{1}{x+y-2}\right)$

$\Rightarrow x + y - 2 = \frac{1}{n};\ x + y = \left(\frac{1+2n}{n}\right)$

43. A, D

Sol. $n(s) = (675n^6 - 2)$

Let k be the middle number

$\therefore$ numbers are $k - 2, k - 1, k, k + 1, k + 2$

$\therefore 3k = \beta^2$ or $k = 3\lambda^2$

also $5k = q^3$

$15\lambda^2 = q^3 \quad \therefore \lambda = 15\mu^3$

$15.15^2.\mu^6 = q^3 \therefore k = 3.225\mu^6 \Rightarrow k = 675\mu^6$

$\Rightarrow k \in \{675, 675.1^6, 675.2^6, \ldots., 675.n^6\}$

$\Rightarrow n(E) = (n + 1) \Rightarrow P(E) = \left(\frac{n+1}{675n^6 - 2}\right)$

44. A, B, D

Sol. $\vec{r}_1 \cdot \vec{r}_2 = |\vec{r}_1| \cdot |\vec{r}_2| \cos\theta \Rightarrow \cos\theta = 1$. Hence, $\vec{r}_1$ and $\vec{r}_2$ are collinear vectors $\Rightarrow \frac{x}{a} = \frac{y}{b} = \frac{z}{c} = \frac{6}{5}$

45. A, D

Sol. When $\cos x < 0$, $x^2 + \cos x - 1 = x^2 - 1 - \cos x \Rightarrow \cos x = 0 \Rightarrow x = \frac{3\pi}{2}$

When $x \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$, $\cos x > 0$, $x^2 + \cos x - 1 = x^2 - 1 + \cos x \Rightarrow \forall\ x \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$

$\Rightarrow x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2}\right)$

46. B, D

Sol. $(x^4 + x^2 - 2) - (a + 1)x(x^2 - 1) = 0,$

$(x^2 + 2)(x^2 - 1) - (a + 1)x(x^2 - 1) = 0$

$(x^2 - 1)(x^2 - (a + 1)x + 2) = 0$

$\Rightarrow x^2 - (a + 1)x + 2 = 0$ have atleast one positive real root

$D \geq 0 \Rightarrow (a + 1)^2 - 8 \geq 0$

$\Rightarrow a \in \left(-\infty, -1 - 2\sqrt{2}\right] \cup \left[2\sqrt{2} - 1, \infty\right)$ and since $\alpha\beta = 2$

Hence, both root, should be positive $\Rightarrow (a + 1) > 0 \Rightarrow a > -1 \Rightarrow a \in \left[2\sqrt{2} - 1, \infty\right)$

47. B, C, D

Sol. α, β, γ are roots of equation $x^3 + ax + a = 0 \Rightarrow \alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \alpha\gamma = a$, $\alpha\beta\gamma = -a$

Also, $\alpha^3 = -a(\alpha + 1)$, $\beta^3 = -a(\beta + 1)$, $\gamma^3 = -a(r + 1) \therefore \frac{\alpha^2}{\beta} + \frac{\beta^2}{\gamma} + \frac{\gamma^2}{\alpha} = -8$

$\Rightarrow \alpha^3\gamma + \beta^3\alpha + \gamma^3\beta + 8\alpha\beta\gamma = 0, -a(\alpha + 1)\gamma - a(\beta + 1)\alpha - a(\gamma + 1)\beta - 8a = 0$

$-a(\alpha\gamma + \gamma + \beta\alpha + \alpha + \gamma\beta + \beta + 8) = 0, a \neq 0 \Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha + (\alpha + \beta + \gamma) + 8 = 0$

$a + 8 = 0 \Rightarrow a = -8 \therefore x^3 - 8x - 8 = 0 \Rightarrow x = -2, 1 \pm \sqrt{5}$

48. A, B, C, D

Sol. $7, g_1, g_2, g_3, g_4, -7$ are in G.P. $\Rightarrow -7 = 7 \cdot r^5 \Rightarrow r = -1 \Rightarrow g_2 = g_4 = 7$ and $g_1 = g_3 = -7$

Hence, equation are $x^2 - 14x + 49 = 0 \Rightarrow (x - 7)^2 = 0$ and $x^2 + 14x + 49 = 0 \Rightarrow (x + 7)^2 = 0$

$\Rightarrow x = \pm 7$ are roots.

SECTION – C

49. 00163.00

Sol. $\Sigma w_r = \Sigma z_r = 3 + 504i$, let $z_r = 3i + k_r(a + ib)$

$\Rightarrow \sum_{r=1}^{5} z_r = 15i + k(a + ib)$ when $k = \Sigma k_r$.

$\Rightarrow b/a = 163$ is the slope of the line L

50. 00000.75

Sol. Will reach at A(2, 2) either in four steps of different combination EENN

$= \frac{4!}{2!2!} = \frac{4 \times 3 \times 2}{2 \times 2}$ = 6 ways

(or) in 6 steps:

EEENNW (or) EENNNS

Where (EENN in any order cannot come in 1st four steps since that it is at point A)

$\therefore$ Number of ways = $2\left(\frac{6!}{3!2!} - \frac{4!}{2!2!} \times 2\right) = 2\left(\frac{6 \times 5 \times 4}{2} - 6 \times 2\right) = 96$

$\therefore$ Probability = $\frac{6}{4^4} + \frac{96}{4^6} = \frac{3}{64}$

51. 00184.00

Sol. If 9^x has leftmost digit as 9 and number of digit is m then 9^{x-1} i.e. $(9x \div 9)$ will have leftmost digit 1 and number of digits remain same m.

Where as if 9^x has leftmost digit other than 9, then and number of digits is m then on division by 9 i.e. in 9^{x-1} number of digit changes to $(m - 1)$.

$\Rightarrow$ every time number of digits increases by 1 by multiplying by 9 then the number was not having left most digit as 9.

Now $9' = 9$ (1 digit number)

9^{4000} has leftmost digit is 9 and number of digits is 3817

$\therefore$ number of digit increases from 1 to 3817

Hence 3816 numbers are not starting with 9 (i.e. leftmost digit is not 9)

and rest $4000 - 3816 = 184$ number are which have left most digit as 9

52. 00364.00

Sol. $f(52, 14) = f(14, 52) = \left(1 + \frac{14}{24}\right).f(14, 38)$

$= \frac{26}{19}\left(1 + \frac{14}{24}\right) f(14,24) = \frac{26}{19} \cdot \frac{19}{12}\left(1 + \frac{14}{10}\right) f(14,10) = 364$

53. 00571.60

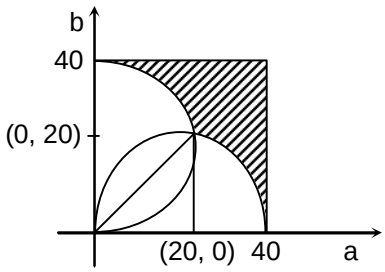
Sol. Let $z = a + ib$

$\Rightarrow 0 < \text{Re}\left(\frac{z}{40}\right) < 1,\ 0 < \text{Im}\left(\frac{z}{40}\right) < 1,$

$0 < \text{Re}\left(\frac{40}{\bar{z}}\right) < 1,\ 0 < \text{Im}\left(\frac{40}{\bar{z}}\right) < 1$

$\Rightarrow 0 < a, b < 40$ and $0 < 40a < a^2 + b^2$ and $0 < 40b < a^2 + b^2$

$\Rightarrow (a - 20)^2 + b^2 > 20^2$ and $a^2 + (b - 20)^2 > 20$



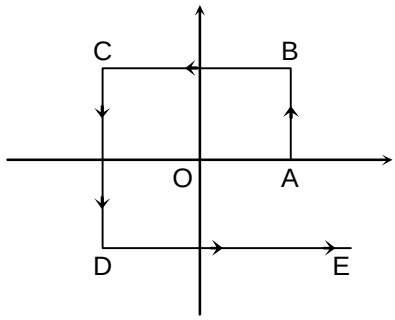
Required area = $40 \times 40 - \pi(20)^2 - 2\left(\frac{1}{4} \times \pi(20)^2 - \frac{1}{2} \times 20 \times 20\right)$

$= 1600 - 400\pi + 200\pi - 400$

$= 1200 - 200\pi = 200(6 - \pi) = 571.6$

54. 00580.00

Sol. From origin all distance parallel to x-axis are



$$\bar{x} = \frac{1^2}{2} - \frac{3^2}{2} + \frac{5^2}{2} - \frac{7^2}{2} + ... + \frac{37^2}{2} - \frac{39^2}{2}$$

$= \frac{1}{2}((-2)(4) + (-2)(12) + (-2)(20) + ... + (-2)(76))$

$= -4(1 + 3 + 5 + ...+19) = -4(100) = -400$

$y = \frac{1}{2}(2^2 - 4^2 + 6^2 - 8^2 + 10^2 - 12^2 + ... + 38^2 - 40^2)$

$= \frac{1}{2}((-2)(6) + (-2)(14) + (-2)(22) + ... + (-2)(78))$

$= -2(3 + 7 + 11 + 15 + ... + 39)$

$= -2\left(\frac{10}{2}(3+39)\right) = -420$

Point will be (–400, –420) its distinct from origin

$d = \sqrt{(400)^2 + (420)^2} = 10\sqrt{1600 + 1764}$

$= 10\sqrt{3364} = 580$ units